

Syed Muhammad Taha Imam

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RESEARCH INTERESTS

Efficient agentic AI for resource-constrained settings: small language models as agents, and deciding with guarantees when an agent should spend another reasoning step, tool call or round of computation. *Foundation from current work:* model compression (structured pruning, early exit, distillation), measuring structural redundancy against proper controls, and conformal risk control for efficiency decisions.

EDUCATION

National University of Sciences & Technology (NUST), Islamabad, Pakistan 2022 – 2026
B.S. in Computer Science, School of Electrical Engineering & Computer Science cGPA: 3.5/4.0
Relevant coursework: Linear Algebra, Probability & Statistics, Machine Learning, Deep Learning. *Honors:* Dean's List (5 semesters).

Cedar College, Pakistan 2020 – 2022
A-Levels: 3 A*, GPA 4.33/4.0

PUBLICATIONS

Peer-Reviewed Conference Papers

- [C1] Tayyab Raza, **Syed Muhammad Taha Imam**, Adrian Ulges, Ulrich Schwanecke, Momina Moetesum, Faisal Shafait. *LiteDoc: Distilling Large Document Models into Efficient Task-Specific Encoders*. In *Document Analysis and Recognition – ICDAR 2026*, Lecture Notes in Computer Science, pp. 418–434. Springer, 2026. [\[DOI\]](#)

Manuscripts (Submitted & In Preparation)

- [M1] **Syed Muhammad Taha Imam**. *Similarity Is Not Importance: On Measuring Representational Redundancy in Wireless Foundation Models*. Manuscript in preparation, 2026. (Full draft with results; available on request.)
- [M2] **Syed Muhammad Taha Imam**. *Certified Early Exit in a Wireless Foundation Model When the Signal-to-Noise Ratio Must Be Estimated*. Manuscript in preparation, 2026. (Full draft with results; available on request.)
- [M3] Nuzhat Akram, **Syed Muhammad Taha Imam**, et al. *Accuracy of Craniometric Features in Gender Estimation Using Machine Learning Algorithms on University of Tennessee (UT) and Howells Datasets*. Submitted, 2026.

RESEARCH EXPERIENCE

Researcher, IPT Lab, SEECS, NUST Jun 2026 – present

- Leading two studies on redundancy and efficient inference in foundation models [M1, M2].
- Tested whether representational similarity (CKA, Procrustes) predicts which transformer blocks can be removed, against untrained-network controls, random removal and a published pruning criterion; similarity-based removal orders largely reproduce an untrained network's ranking ($\rho \geq 0.88$).
- Certified early exit with learn-then-test and exact binomial bounds, plus LoRA adapters (7.3% of encoder weights): 14–39% of encoder compute saved with error $\leq \alpha$ at 95% confidence.
- Showed the guarantee breaks when the routing group (SNR) is estimated rather than known, and that certifying on the estimated groups restores it.

Research Intern, Bradbury Lab (independent research group, remote) Apr 2025 – Jan 2026

- Reviewed literature on Transformer topology and parameter-efficient fine-tuning, focusing on weight-sharing methods.
- Proposed a training-free layer-merging method based on Tucker decomposition to reduce parameter count.
- Analyzed self-attention to assess the feasibility of aligning Query (Q) and Key (K) projections in efficient-by-design architectures.

Undergraduate Research Intern, MachVis Lab, NUST

Oct 2025 – Dec 2025

- Worked on an LLM-based factual-verification framework for the clinical accuracy of generated pathology reports, beyond linguistic metrics (BLEU/ROUGE).
- Curated a challenge set of 100+ gigapixel whole-slide images with real-world artifacts and stain variation to test model robustness.

Undergraduate Research Intern, NUST (remote)

Jun 2025 – Sep 2025

- Combined MedSAM with meta-learning for few-shot segmentation of dental radiographs, improving segmentation of scarce tooth-disease classes by 12%.

TEACHING & LEADERSHIP

- **Teaching Assistant, Linear Algebra**, NUST (Jan 2024 – May 2024): delivered lectures to 40+ undergraduates on applications of linear algebra in machine learning.
- **Volunteer Tutor, CS & Mathematics (O-Levels)** (Jan 2023 – Jun 2023).
- **Executive**, NUST Science Society; **Team Lead**, Google Developer Student Clubs (GDSC), NUST.

INDUSTRY EXPERIENCE

AI Engineer, Brainbox Automations

Dec 2025 – present

- Deployed a safety-guardrailed AI coaching backend on GCP Cloud Run (Dockerized FastAPI, Vertex AI embeddings) serving 300+ minor and adult athletes, with content-moderation guardrails; cut CI/CD deployment time from about 2 minutes to 10 seconds.
- Built SQL-aware agents for a RAG chatbot over 50+ Supabase tables, halving response time.
- Built chat and voice AI (VAPI) for an Australian automation firm: 200+ daily interactions, 5,000+ calls handled, 80% resolved without a human; summarized hour-long call transcripts with parallel map-reduce on AWS Lambda.

Software Engineer (AI) & Team Lead, Epistemy UK

Sep 2025 – Dec 2025

- Built an AI tutoring platform end to end (Nest.js, Next.js, Redis/BullMQ multi-agent queue) with 90%+ test coverage; led a team of two engineers.

Machine Learning Intern, CogniMind AI, NSTP, NUST

Feb 2025 – Apr 2025

- Improved VLM extraction accuracy by 20% and retrieval/inference efficiency by 10% (quantization, HNSW); deployed Dockerized Airflow (5+ DAGs) and CI/CD, cutting deployment time by 30%.

Machine Learning Intern, RapidsAI, SINES, NUST

Sep 2024 – Dec 2024

- Built a RAG chatbot with multi-model query routing, halving API costs.

SELECTED PROJECTS

GPT-2 from Scratch: Pretraining, Efficient Inference and Fine-Tuning

[gpt-2](#)

- Pretrained a GPT-2 language model from scratch in PyTorch on FineWeb-Edu (10B tokens) with FlashAttention.
- Implemented KV caching and speculative decoding for faster autoregressive inference.
- Instruction-tuned the model with supervised fine-tuning (SFT) on the Alpaca dataset.

RAG Tooling: create-rag-app and Agentic-RAG

[create-rag-app](#) | [agentic-rag](#)

- A Python CLI that scaffolds Dockerized RAG apps (13 Jinja2 templates, 3 retrieval strategies), and a LangGraph planner/replan agent evaluated with RAGAS at >95% faithfulness.

HONORS & AWARDS

- Winner, Epistemy week-long hackathon (70+ participants); awarded a paid internship.
- Finalist, VisionRD Hackathon: action-recognition pipeline with a custom backbone.

TECHNICAL SKILLS

Methods	Pruning, early exit, distillation, LoRA/PEFT, CKA/Procrustes, conformal risk control
ML & Systems	Python, PyTorch, CUDA, FlashAttention, HuggingFace, TensorFlow, vLLM, quantization
Engineering	Git, Docker, CI/CD, GCP Cloud Run, AWS Lambda, Airflow, Redis, FastAPI
Certifications	Deep Learning Specialization (DeepLearning.AI)
Languages	English (IELTS 8.0), Urdu (native), German (beginner)